Description of HPGL commands for LP-HFD Laser Projectors

LP-Client 11.40

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Method of data communication
The following instructions are available in direct communication mode with the laser projector (without LPM). All commands are – unless noted – valid in serial and network communication mode, both.

For the serial connection a zero-modem cable has to be used. For the network connection a standard patch cable can be used.

The client software by default waits for a programmable timeout before processing the received data in this operation mode (default timeout is 1500 ms).
In serial mode, the timeout can be suppressed by "embedding" the data in between special control characters `STX` and `ETX` which indicate the start and end of a transmission, respectively. It should be noted, that `STX` and `ETX` represent non-printable, 1-byte control characters: `STX` has a hex value of `0x02` and `ETX` has a hex value of `0x03`. The behavior of `STX/ETX` depends on the setting of the parameter `StxMode` in the WebInterface. When using `STX/ETX` it is recommended NOT to embed each individual command but embed command/data blocks in `STX/ETX` characters.

Settings:
a) Serial Interface

Speed (Baud)	9600 (default)	range: 300 ... 115200 Baud
Data Bits	7	
Stop Bits	1	
Parity	Even	
Flow Control	None	

b) Network Interface (TCP/IP)
Just open a telnet connection to the projector's IP address.

Commands:
All commands should be terminated with `"CR"`, `"LF"` or `";"` to identify exactly one command. Combinations are allowed, commands are not case sensitive and blanks are ignored. For a better readability, however, the usage of blanks is recommended.

Range of values:
Values can be entered as floating point numbers by using `"."` ("point") as decimal point. The delimiter `","` ("comma") is used to separate the individual values. The place after the decimal point should be rounded to 2 digits.

PA (plot absolute)
Plot absolute draws a line from the last known coordinate to the new coordinate with PA x, y, z, where x, y and z are absolute values related to the origin of the coordinate system. The line is drawn visibly after a PD (PenDown) instruction or invisibly after a PU (PenUp instruction).

Notation of the instruction:
PA Xvalue , Yvalue [,Zvalue] ;
PA Xvalue1 , Yvalue1 , Xvalue2 , Yvalue2 , Xvalue n , Yvalue n

PR (plot relative)
Plot relative draws a line from the last known coordinate to the new coordinate with PR x, y, z, where x, y and z are values relative to the last used coordinate. The line is drawn visibly after a PD (PenDown) instruction or invisibly after a PU (PenUp instruction).

Notation of the instruction:
PR Xvalue , Yvalue [,Zvalue] ;
PR Xvalue1 , Yvalue1 , Xvalue2 , Yvalue2 , Xvalue n , Yvalue n

PL (poly line)
Polyline acts almost identical to PA. However PL is a special command in LPM and the projector, in order to emulate a polyline with special relevance for LPM click. Polyline draws a line from the last known coordinate to the coordinates programmed with PL x, y relative to the program zero point. The line is drawn visibly after a PD (PenDown) instruction or invisibly after a PU (PenUp instruction).

Notation of the instruction:
PL Xvalue , Yvalue ;
PL Xvalue1 , Yvalue1 , Xvalue2 , Yvalue2 , Xvalue n , Yvalue n

PD (pen down)
Change drawing mode to visible. The following coordinates are drawn visibly. Depending on the previous instruction, the coordinate is processed absolutely or relatively to the last programmed coordinate.

Notation of the instruction:
PD;
PD Xvalue , Yvalue ;

PU (pen up)
Change drawing mode to invisible. The following coordinates are drawn invisibly. Depending on the previous instruction, the coordinate is processed absolutely or relatively to the last programmed coordinate.

Notation of the instruction:
PU;
PU Xvalue , Yvalue ;

SP (select pen)
The SP command sets the pen color for the following coordinates. The pen color will remain unchanged until the next SP command is received.

Note:
It is recommended to send a PU; command before and a PD; command after the SP command.

Notation of the instruction:
SP n;

CI (circle)
Draws a circle with the radius r in the xy plane at the current height. Center point is the last programmed coordinate.

Notation of the instruction:
CI r;

AA (arc absolute)
Draws an arc from the last programmed coordinate, around the center point x,y with the angle j in degrees.
The angle can be positive or negative (negative is clockwise).

Notation of the instruction:
AA x, y, j;

AR (arc relative)
Draws an arc from the last programmed coordinate, around the center point x,y with the angle j in degrees. The coordinate for the center is relative from the last known coordinate.
The angle value is absolute and can be positive or negative (negative is clockwise).

Notation of the instruction:
AR x, y, j;

ZH (projection height)
All coordinates following a ZH command will be shown at the programmed height specified by the command. The height offset is relative to the Laser Projector's zero height plane. A positive ZH is calculated towards the Laser Projector.

Notation of the instruction:
ZH h; h in mm

LM (line multiple)
This command draws multiple lines parallel to the x axis. The lines are identical in length and at the same projection height.

Notation of the instruction:
LM Height, Length, Y1 [, Y2 [.. ,Yn]] ;
Height: Height of the projection plane [1/10mm]
Length: Length of the projected line(s) [1/10mm]
Y1 .. Yn: Position of the individual lines relative to zero [1/10 mm]
(maximum number of lines: 32)

ZF (projection OFF)
Stop the projection temporarily. A projection can be re-started by using the command ZN. If new projection data is sent, the projection will also be started.

Notation of the instruction:
ZF;

ZN (projection ON)
Start a projection which has been stopped with the ZF command.

Note :
In case of a failed drift compensation, the projection stops. The ZN command re-starts the projection as it was before the drift compensation.

Notation of the instruction:
ZN;

ZD (drift compensation)
Drift compensation eliminates thermal and electrical drift.

Note :
in case of no success of the ZD command, the projector will NOT start the former activated projection. (See ZN)

Notation of the instruction:
ZD;

ZD n dx dy (change the DAP position)
The ZD command has an extension to tweak the projectors accuracy. The drift compensation points are required for long term stability and accuracy of a laser projector.

Notation of the instruction:
ZDnvalue dXvalue dYvalue ;
n is the drift compensation point to be modified
dx and dy are the offset values
ZD2 100,500;

Note: The tweaking is only temporary unless the values are stored with the ZDW command.
For an effective usage of this function, the DAPs should be at the field borders.

ZDW (write changes of ZD)
Make the changes made with command ZD permanent.

Notation of the instruction:
ZDW;

ZP (select pen)
Select one or more pens for the projection.

Notation of the instruction:
ZP value[,value];
ZP 1; Select only pen 1.
ZP 2,3,4; Select pens 2, 3 and 4.

ZA (rotation and offset around zero)
The following coordinates will get an offset and a rotation. A previously programmed offset will be overwritten.

Notation of the instruction:
ZA Xvalue , Yvalue , Rvalue ;
ZA 100,500,56; The drawing will be moved to X=100, Y=500mm and rotated with 56 deg

ZO (offset)
The drawing will get an offset. The offset will be overwritten by a ZA command or the next valid ZO command.

Notation of the instruction:
ZO Xvalue , Yvalue , Zvalue ;
ZO 100,500; The Drawing will be moved to X=100 and Y=500mm.

IN (initialize)
Initializes the Projector. This command does switch on the projection, offset, pens and rotation are not influenced.

Note: The behavior of this command depends on the setting of the parameter `Init Compatibility` in the Web-Interface.

LB (text label)
Project a text. The text to displayed is delimited from LB to the end character (see DT), the next semicolon, LF or CR. The maximum length of the text is 100 characters.

Note: A semicolon cannot be displayed by the projector.

Notation of the instruction:
LB TEXT DT ;
Value Range:
Text is the string, to project
DT is the terminator for the string

DT (string terminator)
A text shown by LB can contain a text terminator. Default is 0x03 (ETX)

Notation of the instruction:
DT c ;
Value Range:
c has to be a valid ASCII character (no control character!)

SI (character size)
Set the size of the text characters. Default is 20 mm (in xy plane).

Notation of the instruction:
SI Xvalue , Yvalue ;
Range: in mm

SR (character size relative)
Relative changes of the character size in percent.

Notation of the instruction:
SI Xvalue , Yvalue ;
Range: 1..1000 percent

DI (orientation of the label)
The orientation of the label correlates with the user coordinate system. The rotation is calculated for the unit vector (adjacent leg/ opposite leg in the direction of the arc tangents.

Notation of the instruction:
DI opp,adj ;
value range: 0..1
opp is the opposite leg
adj is the adjacent leg

Example :
rotate to 90° DI 0,1;
rotate to 180° DI -1,0;
rotate to 30° DI 0.866,0.5;

DR (orientation of the label relative)
Orientation of the label relative (for details see DI).

DF (reset font setting)
This command resets the font settings to the default values.

Notation of the instruction:
DF ;

ZM (reset PIR)
This commands resets a triggered PIR. This is necessary to re-enable a stopped projection.

Notation of the instruction:
ZM;

ZS (select filename)
This commands selects a projection file (.plt) on the projector and starts the projection. This command must be enabled with the parameter `Read Filenames from Serialport or Network` in the WebInterface. For details there is a separate document available.

Notation of the instruction:
ZS filename;

Commands with response
The following commands return a response to the sender.

OI (read basic information)
This command reads basic information from the projector:
- client version
- serial number
- operating time over 40 degrees Celsius [seconds]
- run time of source and galvos [seconds]

Notation of the instruction:
OI

Response:
Version: LW<version>
Serial Number: <serialnumber>
Time >40 C: <time40>
Run Time: <runtime>

Example:
Version: LW1138
Serial Number: S911000040
Time >40 C: 0
Run Time: 4698

?Z (calibration status)
This command returns the status of the last point search.

Notation of the instruction:
?Z

Response:
<code>

where <code> is one of the following values:
+0 point search has not been triggered
+9 point search was successful
-1 point 1 not found
-2 point 2 not found
-3 point 3 not found
-4 point 4 not found

New commands for compatibility
The following commands are active temporary. After restarting the projector, they are reset to default values.

MU (move origin)
Set a new projector origin.

Notation of the instruction:
MU Xvalue , Yvalue , Zvalue ;
MU 100,500,56; The origin is set to X=100 mm, Y=500 mm and Z=56 mm according to the calibration. A later programmed ZO or ZA will use this offset.

MB245 (select one plot buffer direct)
The commands sets the plot buffer to be used. All following commands use this selected plot buffer.

Notation of the instruction:
MB245, Buffer ;

MB246 (delete one plot buffer)
The command deletes a currently active plot buffer and updates the projection immediately if at least one active buffer remains; the last (only) visible buffer will be deleted immediately, the projection, however, will be updated when new data is sent to a plot buffer (the buffer is still visible until then).

Notation of the instruction:
MB246, Buffer;

MB247 (delete all plot buffers)
The command deletes all plot buffers. The deletion will be executed immediately, however, the projection is not upated. The projection will be updated when new data is sent to a plot buffer.
To switch off the projection immediately, consider using the ZF; command to disable the complete projection.

Notation of the instruction:
MB247;

MB241 (address only one projector)
Following commands will be processed only by the addressed projector.

Notation of the instruction:
MB241, Slave;
Range: Slave A..P

MB242 (select one projection field)

A field is a set of information created by the calibration. A projector can have up to 16 fields. This command activates the selected field. The field must exist, means it has to be created during the calibration.

Offset, rotation and origin reset to zero.
Notation of the instruction:
MB242, field;
Range: 1 ..16

MB1 (set projection timer) [caution!]
The projection timer defines the update period/delay between two consecutive points in the output buffer.

Notation of the instruction:
MB1, Timer;
Range: 20 .. 150

MB2 (set step length for PU) [caution!]
For the laser picture the laser projector divides the line into segments (steps). For each TimerTick a new step is calculated. A step length that is too large decreases the visibility and the accuracy of the laser projection, a step length that is too small can cause a flicker projection. Special attention is on laser security.

Notation of the instruction:
MB2, Step length;
Range: 1 .. 50

MB3 (set step length for PD) [caution!]

see MB2

Notation of the instruction:
MB3, Step length;
Range: 1 .. 30

MB4 (corner delay for PU) [caution!]
During projection, the projector has to stop at edges. This delay/stop time is defined in TimerTicks * value, and is programmed by MB4. A value too long slows down the speed of the projection down while a value too short makes corners “round”.

Notation of the instruction:
MB4, DelayUp;
Range: 3 .. 20

MB5 (corner delay for PD) [caution!]
see MB4
Notation of the instruction:
MB4, DelayDown;
Range: 3.. 20

MB6 (set timeout value)
This command sets the timeout value for the communication (in ms).

Notation of the instruction:
MB6,TimeOut;
Range: 10 … 60000 [ms]
Default: 1500 [ms]

MB15 (clipping boundary Xmin)
The clipping border can be adjusted to project only a smaller part/section of the projection.

Note:
Ensure that the values are inside the projection range. The Clipping area refers to the currently active calibration type (z-Easy, working plane, 3D-Setup).

Notation of the instruction:
MB15, Clipping;
Range : in mm

MB16 (clipping boundary Ymin)
see MB15

MB17 (clipping boundary Xmax)
see MB15

MB18 (clipping boundary Ymax)
see MB15

!!! IMPORTANT NOTE !!!

All commands marked with **[caution!]** should not be used unless there are reasons to do so! On projector startup the default values are loaded. Hence, if these commands are not used – which is highly recommended – the default values are used.
Be aware that changing theses parameters may have influence on the lifetime of the projector’s galvos. Hardware defects caused by wrong setting of these parameters will not be covered by the warranty!